Understanding Rule Discovery in Data Validation

An Information Theory Approach

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Analysis for tape-whisperer Project

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Abstract

This document explains how to figure out the number of "rules" or patterns hidden in a spreadsheet of financial data. We use ideas from **information theory**[A1]—the mathematics of measuring information—to answer questions like: How much "surprise" is in each cell? How many rules could possibly exist? And how many are actually useful? We find that while *trillions* of rules are theoretically possible, only about 3,000–5,000 are practically useful.

Note: Every time you see a colored term like **this**[A1], it means there's an explanation in the Appendix. Click it or flip to the back!

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1 What Problem Are We Solving?

Imagine you work at a bank, and every day you receive spreadsheets containing information about thousands of loans. Each spreadsheet has:

- **224 columns** (like "Loan Amount", "Interest Rate", "Borrower State", etc.)
- **Thousands of rows** (one row per loan)

Your job is to check if the data is correct. But here's the challenge: the rules for "correct" are complicated!

Examples of Hidden Rules

- If the loan type is "FHA", then the down payment must be at least 3.5%
- If the property is in California, the tax rate must be between 0.7% and 1.2%
- If the borrower is self-employed, additional income documents are required

Our Goal: Automatically discover these rules from historical data, then use them to find errors in new spreadsheets.

The Big Questions:

1. How much "information" is in each cell of the spreadsheet?
2. How many possible rules could there be?
3. How do we find the useful rules without drowning in useless ones?

2 Measuring Information: Entropy

2.1 What is Entropy?

Entropy[A2] is a way to measure **surprise** or **uncertainty**.

Think of it this way: if someone tells you "the sky is blue," you're not surprised—you already knew that. But if they tell you "it's snowing in Miami in July," you're very surprised!

Key Idea

Entropy measures how surprised you'd be, on average, when you learn something.

- High entropy = lots of surprise = lots of uncertainty = hard to predict
- Low entropy = little surprise = very predictable

2.2 The Entropy Formula

The formula for entropy was invented by Claude Shannon in 1948:

$$H(X) = -\sum_{i=1}^{n} p(x_i) \times \log_2(p(x_i)) \tag{1}$$

Let's break this down piece by piece:

- $H(X)$[A3] — The entropy of variable X, measured in **bits**[A4]

- $\sum$[A5] — The summation symbol ("add up all of these")
- $p(x_i)$[A6] — The probability of outcome x_i happening
- $\log_2$[A7] — The logarithm base 2 (related to binary/computer math)
- The negative sign at the front makes the answer positive (since $\log_2$ of a fraction is negative)

2.3 A Simple Example: Coin Flip

Let's calculate the entropy of a fair coin flip.

Setup:

- Two possible outcomes: Heads or Tails
- Each has probability $p = 0.5$ (50%)

Calculation:

$$\begin{aligned} H(\text{coin}) &= -[p(\text{heads}) \times \log_2(p(\text{heads})) + p(\text{tails}) \times \log_2(p(\text{tails}))] && (2)\\ &= -[0.5 \times \log_2(0.5) + 0.5 \times \log_2(0.5)] && (3)\\ &= -[0.5 \times (-1) + 0.5 \times (-1)] && (4)\\ &= -[-0.5 + (-0.5)] && (5)\\ &= -[-1] && (6)\\ &= 1 \text{ bit} && (7) \end{aligned}$$

What does "1 bit" mean?

One bit is the amount of information needed to answer a single yes/no question. A fair coin flip has exactly 1 bit of entropy because you need exactly one yes/no question to determine the outcome: "Is it heads?"

2.4 Entropy for a Spreadsheet Cell

Now let's apply this to our spreadsheet problem. Consider a cell that can contain one of three types of data:

1. Text (like a name or code)
2. A number (like a dollar amount)
3. A date (like "03/15/2024")

If each type is equally likely (probability $= \frac{1}{3}$ each):

$$\begin{aligned} H(\text{cell type}) &= -3 \times \frac{1}{3} \times \log_2\left(\frac{1}{3}\right) && (8)\\ &= -\log_2\left(\frac{1}{3}\right) && (9)\\ &= \log_2(3) && (10)\\ &\approx 1.585 \text{ bits} && (11) \end{aligned}$$

Key Idea

When something is completely predictable (100% one outcome), entropy is **zero**. When all outcomes are equally likely, entropy is at its **maximum**.

Table 1: Entropy for different scenarios

Scenario	Probabilities	Entropy
Equal chance of 3 types	(33%, 33%, 33%)	1.585 bits
Mostly numbers	(60%, 30%, 10%)	1.295 bits
Almost always numbers	(90%, 5%, 5%)	0.569 bits
Always the same type	(100%, 0%, 0%)	0 bits

3 Entropy of an Entire Spreadsheet

3.1 Adding Up All the Cells

Our spreadsheet has:

- 100 rows
- 224 columns
- Total cells: $100 \times 224 = 22,400$ cells

If each cell were **independent**[A8] (not related to other cells), the total entropy would simply be:

$$H_{\text{total}} = (\text{number of cells}) \times H_{\text{per cell}} \tag{12}$$

Using our 3-type example:

$$H_{\text{total}} = 22,400 \times 1.585 \tag{13}$$
$$\approx 35,502 \text{ bits} \tag{14}$$

This means it would take about 35,502 yes/no questions to completely describe the spreadsheet—if every cell were random and unpredictable.

3.2 But Real Data Has Structure!

Here's the key insight: **real spreadsheets aren't random**. They have structure:

- Column A is *always* a date
- Column B is *always* a loan ID
- If the loan type is "FHA", certain fields *must* have specific values

This structure creates **dependencies**[A9] between cells. When cells are dependent, knowing one cell tells you something about another cell.

Example of Dependency

If I tell you "the loan type is FHA", you can immediately predict:

- The minimum down payment (3.5%)
- The required insurance type (MIP)
- The maximum loan amount (varies by county)

These predictions reduce your uncertainty—and therefore reduce entropy!

3.3 Conditional Entropy

Conditional entropy[A10] measures how much uncertainty remains *after* you learn something else.

We write it as:

$$H(B|A) \tag{15}$$

This reads as: "The entropy of B, *given* that we know A."

Key Idea

Rules are relationships where knowing A reduces uncertainty about B.
Mathematically: $H(B|A) < H(B)$
In words: "If knowing the loan type helps us predict the interest rate, then there's a rule connecting them."

4 What is a "Rule"?

4.1 Association Rules

An **association rule**[A11] is a statement of the form:

$$\boxed{\textbf{IF } [\text{condition}] \textbf{ THEN } [\text{consequence}]}$$

Examples:

- IF loan_type = "FHA" THEN down_payment $\geq 3.5\%$
- IF state = "CA" AND property_type = "Condo" THEN HOA_required = "Yes"
- IF income_verified = "No" THEN max_LTV $\leq 80\%$

4.2 Parts of a Rule

Every rule has two parts:

1. **Antecedent** (the "IF" part): The condition(s) that must be true
2. **Consequent** (the "THEN" part): What follows when the condition is true

The number of conditions in the antecedent is called k[A12]:

- $k = 1$: One condition (e.g., IF loan_type = "FHA")
- $k = 2$: Two conditions (e.g., IF loan_type = "FHA" AND state = "CA")
- $k = 3$: Three conditions (and so on...)

5 How Many Rules Could Exist?

5.1 Counting Rule Structures

First, let's count how many different rule "shapes" are possible, ignoring specific values.

With 224 columns, we need to choose:

1. Which columns form the antecedent (the "IF" conditions)

2. Which column is the consequent (the "THEN" result)

The formula uses **combinations**[A13]:

$$N_{\text{structures}}(n, k) = n \times \binom{n-1}{k} \tag{16}$$

Where:

- n = number of columns (224)
- k = number of antecedent columns
- $\binom{n-1}{k}$ = "n-1 choose k" = ways to pick k items from $n-1$

Table 2: Number of rule structures by antecedent size

Antecedent Size (k)	**Rule Structures**	**In Powers of 10**
1	49,952	$\approx 5 \times 10^4$
2	5,578,496	$\approx 5.6 \times 10^6$
3	414,315,776	$\approx 4.1 \times 10^8$
4	23,063,051,456	$\approx 2.3 \times 10^{10}$
5	1,024,377,696,256	$\approx 1.0 \times 10^{12}$

5.2 Adding Specific Values

But wait—we haven't counted the specific *values* yet!

The rule "IF loan_type = FHA" is different from "IF loan_type = VA".

If each column has about 20 possible values on average, then for each rule structure, we multiply by the number of value combinations:

$$N_{\text{full}} = N_{\text{structures}} \times (\text{average values per column})^k \tag{17}$$

For $k = 2$ (two-condition rules):

$$\begin{aligned} N_{\text{full}} &= 5,578,496 \times 20^2 && (18)\\ &= 5,578,496 \times 400 && (19)\\ &\approx 2.2 \text{ billion rules} && (20) \end{aligned}$$

For $k = 3$:

$$\begin{aligned} N_{\text{full}} &= 414,315,776 \times 20^3 && (21)\\ &\approx 3.3 \text{ trillion rules} && (22) \end{aligned}$$

The Rule Explosion Problem

With just 224 columns and rules up to 3 conditions, there are over **3 trillion** possible rules!

Even if a computer could check 1 million rules per second, it would take over 100 years to check them all.

This is why we need smart ways to filter down to useful rules.

6 Measuring Rule Quality

Not all rules are equally useful. We need ways to measure rule quality.

6.1 Support: How Often Does This Rule Apply?

Support[A14] measures how frequently a rule's conditions appear in the data.

$$\text{Support}(A \Rightarrow B) = \frac{\text{Number of rows where A AND B are true}}{\text{Total number of rows}} \tag{23}$$

Example: If 15,000 out of 100,000 loans are FHA loans with valid down payments:

$$\text{Support} = \frac{15,000}{100,000} = 0.15 = 15\%$$

Why Support Matters

A rule with very low support (say, 0.001%) might only apply to 1 loan out of 100,000. Even if it's correct, it's not very useful because it rarely applies.
Typical minimum: We often require at least 0.1% support (100 rows out of 100,000).

6.2 Confidence: When It Applies, How Often Is It Right?

Confidence[A15] measures how reliable a rule is when its conditions are met.

$$\text{Confidence}(A \Rightarrow B) = \frac{P(A \text{ and } B)}{P(A)} = \frac{\text{Support}(A \Rightarrow B)}{\text{Support}(A)} \tag{24}$$

Example:

- 15% of all loans are FHA with valid down payments
- 16% of all loans are FHA (regardless of down payment)

$$\text{Confidence} = \frac{0.15}{0.16} = 0.9375 = 93.75\%$$

This means: "When a loan is FHA, 93.75% of the time it has a valid down payment."

6.3 Lift: Is This Rule Better Than Guessing?

Lift[A16] compares the rule to random chance.

$$\text{Lift}(A \Rightarrow B) = \frac{\text{Confidence}(A \Rightarrow B)}{P(B)} \tag{25}$$

- **Lift = 1**: The rule is no better than random chance
- **Lift > 1**: The rule is better than guessing (useful!)
- **Lift < 1**: The rule is worse than guessing (anti-pattern)

Example:

- Confidence = 93.75%
- Overall, 85% of all loans have valid down payments

$$\text{Lift} = \frac{0.9375}{0.85} = 1.10$$

This rule is 10% better than just guessing "valid down payment" for everyone.

7 Information Theory Meets Rule Quality

7.1 Rules Reduce Uncertainty

Here's the deep connection: **good rules reduce entropy**.

When you learn the loan type, you become less uncertain about other fields. This reduction in uncertainty is called **information gain**[A17]:

$$\text{Information Gain} = H(B) - H(B|A) \tag{26}$$

Where:

- $H(B)$ = entropy of B before knowing A (original uncertainty)
- $H(B|A)$ = entropy of B after knowing A (remaining uncertainty)

7.2 A Rule's Value

We can combine support and information gain to measure a rule's total value:

$$\text{Value(rule)} = \text{Support} \times \text{Information Gain} \tag{27}$$

This represents: "How much total uncertainty does this rule eliminate across all the data?"

Key Idea

Rule discovery is about finding the rules with the highest value.
A rule is valuable if:

1. It applies frequently (high support)
2. It tells you a lot when it applies (high information gain)

7.3 Pointwise Mutual Information (PMI)

PMI[A18] is another way to measure how much two things are related:

$$\text{PMI}(A, B) = \log_2 \frac{P(A \text{ and } B)}{P(A) \times P(B)} \tag{28}$$

- PMI > 0: A and B occur together more than expected (positive relationship)
- PMI $= 0$: A and B are independent (no relationship)
- PMI < 0: A and B occur together less than expected (negative relationship)

Connection to Lift: PMI $= \log_2(\text{Lift})$

8 The Description Length Principle

8.1 Minimum Description Length (MDL)

Here's a powerful idea from information theory: **MDL**[A19] says the best model is the one that provides the shortest total description of:

1. The model itself (the rules)
2. The data given the model (exceptions to the rules)

$$\text{Total Length} = L(\text{Rules}) + L(\text{Data}|\text{Rules}) \tag{29}$$

8.2 Why This Helps

Imagine two extremes:

1. **No rules:** $L(\text{Rules}) = 0$, but $L(\text{Data})$ is huge (every cell is a surprise)
2. **Too many rules:** $L(\text{Rules})$ is huge, $L(\text{Data}|\text{Rules})$ is small

The optimal point is somewhere in between—enough rules to capture real patterns, but not so many that we're memorizing noise.

8.3 Cost of a Rule

We can calculate how many bits it takes to describe a rule:

$$L(\text{rule}) = \underbrace{\log_2 \binom{n}{k+1}}_{\text{which columns}} + \underbrace{k \times \log_2 c}_{\text{antecedent values}} + \underbrace{\log_2 c}_{\text{consequent value}} \tag{30}$$

Where:

- n = total columns (224)
- k = number of antecedent conditions
- c = average values per column (about 20)

Table 3: Approximate encoding cost per rule

Antecedent Size (k)	**Bits to Encode Rule**
1	$\approx$ 24 bits
2	$\approx$ 32 bits
3	$\approx$ 41 bits
4	$\approx$ 49 bits

8.4 When Is a Rule Worth Adding?

A rule is worth adding if it saves more bits than it costs:

$$\underbrace{L(\text{rule})}_{\text{cost}} < \underbrace{\text{Support} \times \text{Rows} \times \text{Info Gain}}_{\text{savings}} \tag{31}$$

Example: For a $k = 2$ rule (cost $\approx$ 32 bits) with 100,000 rows:

- At 1% support (1,000 rows): need ≥ 0.032 bits/row gain
- At 10% support (10,000 rows): need ≥ 0.0032 bits/row gain

9 Practical Filtering: From Trillions to Thousands

9.1 The Filtering Cascade

We start with trillions of possible rules and filter down through several stages:

1. **Support filter**: Remove rules that apply to too few rows

2. **Confidence filter**: Remove rules that aren't reliable enough
3. **Lift filter**: Remove rules that aren't better than guessing
4. **Information gain filter**: Remove rules that don't reduce uncertainty

9.2 Typical Thresholds

Table 4: Quality thresholds for useful rules

Metric	Minimum Value	What It Means
Support	0.1%	At least 100 rows (of 100K)
Confidence	80%	Right 4 out of 5 times
Lift	1.5	50% better than guessing
Information Gain	0.05 bits	Meaningful reduction in uncertainty

9.3 The Final Numbers

Table 5: How filtering reduces the rule space

Stage	$k = 1$	$k = 2$	$k = 3$
All possible structures	49,952	5,600,000	414,000,000
After support filter	2,500	28,000	104,000
After confidence filter	250	2,800	10,400
After info gain filter	**125**	**1,400**	**5,200**

The Bottom Line

Theoretical maximum: Over 3 trillion possible rules
After quality filtering: About 3,000–5,000 useful rules
Compression ratio: We keep only 1 rule for every billion possibilities!

10 Summary

10.1 Key Takeaways

1. **Entropy measures uncertainty.** A coin flip has 1 bit of entropy. Spreadsheet cells have 1–20 bits depending on the field type.
2. **Rules are patterns that reduce entropy.** When knowing column A tells you something about column B, there's a rule connecting them.
3. **The rule space is astronomically large.** With 224 columns, there are trillions of possible rules.
4. **Quality filtering is essential.** Support, confidence, lift, and information gain help us find the needles in the haystack.
5. **MDL provides a principled framework.** A rule is worth keeping if it saves more information than it costs to describe.

6. **In practice, we find 3,000–5,000 useful rules.** This is manageable for a computer system to store and apply.

10.2 For the tape-whisperer System

This analysis tells us:

- We can expect to discover several thousand useful validation rules
- These rules can be stored in a few megabytes of YAML files
- The FP-Growth (Frequent Pattern) algorithm can find them in minutes, not years
- The rules will cover the major patterns while ignoring noise

Appendices: Glossary of Terms

Each term below is referenced in the main text. They're organized alphabetically by their reference code.

A1: Information Theory

Information theory is a branch of mathematics that studies how to measure, store, and transmit information. It was invented by Claude Shannon in 1948.

Everyday analogy: Think about texting. Some messages are longer than others because they contain more information. Information theory gives us precise ways to measure "how much" information is in a message.

Key insight: Information is related to surprise. "The sun rose today" contains little information (not surprising). "Aliens landed in Paris" contains a lot of information (very surprising!).

A2: Entropy

Entropy is a number that measures uncertainty or randomness.

Simple examples:

- A fair coin: entropy = 1 bit (maximum uncertainty for 2 outcomes)
- A die with all sides showing 6: entropy = 0 bits (no uncertainty)
- A fair 6-sided die: entropy = 2.58 bits

Why "entropy"? The term comes from physics, where it measures disorder. Shannon borrowed it because mathematical properties are similar.

Rule of thumb: For n equally likely outcomes, entropy = $\log_2(n)$ bits.

A3: $H(X)$ — Entropy Notation

When we write $H(X)$, we mean "the entropy of X."

- H is the standard letter used for entropy (named after physicist Ludwig Boltzmann's H-theorem)
- X is whatever we're measuring (a coin flip, a spreadsheet cell, etc.)

Examples:

- $H(\text{coin flip}) = 1$ bit
- $H(\text{weather tomorrow})$ might be 2 bits (sunny, cloudy, rainy, snowy)
- $H(\text{loan type})$ might be 3 bits if there are 8 equally likely types

A4: Bits

A **bit** is the smallest unit of information.

What it represents: The answer to one yes/no question.

Examples:

- "Is the number greater than 50?" — answering this gives you 1 bit
- To identify a card from a deck of 52, you need about 6 bits ($\log_2(52) \approx 5.7$)
- A byte = 8 bits, which is enough to represent any letter, number, or symbol

In computers: Everything is stored in bits (0s and 1s). A 1-megabyte file contains about 8 million bits.

A5: $\sum$ — Summation

The symbol $\sum$ (capital Greek letter sigma) means "add up all of these."

How to read it:

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + x_3 + \cdots + x_n$$

Example: The sum of numbers 1 through 5:

$$\sum_{i=1}^{5} i = 1 + 2 + 3 + 4 + 5 = 15$$

In entropy formula: We add up the "surprise" contribution from each possible outcome.

A6: $p(x_i)$ — Probability

$p(x_i)$ means "the probability that outcome x_i happens."

Properties of probability:

- Always between 0 and 1 (or 0% and 100%)
- All probabilities must add up to 1 (something has to happen!)

Examples:

- For a fair coin: $p(\text{heads}) = 0.5$
- For a fair die: $p(\text{rolling a 6}) = \frac{1}{6} \approx 0.167$
- For loan types: $p(\text{FHA}) = 0.15$ means 15% of loans are FHA

A7: $\log_2$ — Logarithm Base 2

$\log_2(x)$ asks: "2 raised to what power gives x?"

Examples:

- $\log_2(2) = 1$ because $2^1 = 2$
- $\log_2(4) = 2$ because $2^2 = 4$
- $\log_2(8) = 3$ because $2^3 = 8$
- $\log_2(1) = 0$ because $2^0 = 1$
- $\log_2(0.5) = -1$ because $2^{-1} = 0.5$

Why base 2? Because computers use binary (0s and 1s). This makes the answer come out in bits.

Key property: $\log_2(a \times b) = \log_2(a) + \log_2(b)$

A8: Independence

Two things are **independent** if knowing one tells you nothing about the other.

Independent examples:

- Flipping two different coins — the first flip doesn't affect the second
- Your birthday and tomorrow's weather

NOT independent (dependent):

- Height and shoe size (taller people tend to have bigger feet)
- Loan type and required down payment (FHA requires specific minimums)

Mathematical test: A and B are independent if $P(A \text{ and } B) = P(A) \times P(B)$

A9: Dependencies

A **dependency** exists when one variable provides information about another.

In spreadsheets: Dependencies are the basis of validation rules. If Column A's value constrains what's valid in Column B, there's a dependency.

Types of dependencies:

- *Functional*: B is completely determined by A (like birth date → age)
- *Partial*: A gives partial information about B (like zip code → likely state)
- *Statistical*: A and B tend to co-occur (like high income → high credit score)

A10: Conditional Entropy

Conditional entropy $H(B|A)$ measures uncertainty about B *after* you learn A.

Key insight: $H(B|A) \leq H(B)$ always. Knowing something never increases uncertainty!

Example:

- $H(\text{interest rate}) = 4$ bits (many possible rates)
- $H(\text{interest rate}|\text{loan type}) = 2$ bits (knowing loan type narrows it down)
- Information gained $= 4 - 2 = 2$ bits

A11: Association Rule

An **association rule** is a pattern of the form: IF [conditions] THEN [result].

Famous example from retail: "IF customer buys diapers THEN customer likely buys beer"

(This famous pattern was supposedly discovered in supermarket data!)

In data validation: "IF loan_type = 'FHA' AND property_type = 'Condo' THEN HOA_status must be 'Active"'

A12: k — Antecedent Size

k represents the number of conditions in the "IF" part of a rule.

- $k = 1$: Simple rule with one condition
- $k = 2$: Rule with two conditions combined with AND
- $k = 3$: Three conditions, etc.

Trade-off:

- Higher k = more specific rules (potentially more accurate)
- Higher k = exponentially more possible rules (harder to find)
- Higher k = rules apply to fewer rows (lower support)

In practice: Most useful rules have k between 1 and 3.

A13: Combinations

$\binom{n}{k}$ (read "n choose k") counts how many ways to select k items from n items when order doesn't matter.

Formula:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

where $n!$ ("n factorial") $= n \times (n-1) \times (n-2) \times \cdots \times 1$

Examples:

- $\binom{5}{2} = 10$ (ways to choose 2 items from 5)
- $\binom{52}{5} = 2,598,960$ (possible 5-card poker hands)
- $\binom{224}{3} = 1,847,424$ (ways to choose 3 columns from 224)

A14: Support

Support is the frequency of a rule in the data—what fraction of rows match the rule's conditions AND conclusion.

Formula:

$$\text{Support} = \frac{\text{rows matching both IF and THEN}}{\text{total rows}}$$

Why it matters: A rule with 0.0001% support only applies to 1 in a million rows. Even if it's correct, it's not very useful.

Typical threshold: 0.1% to 1% minimum support

A15: Confidence

Confidence measures how often the rule is correct when its conditions are met.

Formula:

$$\text{Confidence} = \frac{\text{rows matching both IF and THEN}}{\text{rows matching IF}}$$

Example:

- 1,000 loans are FHA with valid down payment
- 1,100 loans are FHA total
- Confidence $= \frac{1000}{1100} = 90.9\%$

Typical threshold: 80% or higher

A16: Lift

Lift measures how much better a rule is compared to random chance.

Formula:

$$\text{Lift} = \frac{\text{Confidence}}{\text{Probability of conclusion happening anyway}}$$

Interpretation:

- Lift = 1: No better than guessing
- Lift = 2: Twice as good as guessing
- Lift = 0.5: Half as good as guessing (anti-pattern)

A17: Information Gain

Information gain is the reduction in entropy when you learn something new.

Formula:

$$\text{Information Gain} = H(\text{before}) - H(\text{after})$$

Example:

- Before knowing loan type: 5 bits of uncertainty about rate
- After knowing loan type: 2 bits of uncertainty about rate
- Information gain: 3 bits

Connection to rules: A good rule has high information gain—knowing the IF part tells you a lot about the THEN part.

A18: PMI — Pointwise Mutual Information

PMI measures how much more (or less) often two things occur together compared to if they were independent.

Formula:

$$\text{PMI}(A, B) = \log_2 \frac{P(A \text{ and } B)}{P(A) \times P(B)}$$

Interpretation:

- PMI > 0: A and B occur together more than expected
- PMI = 0: A and B are independent
- PMI < 0: A and B avoid each other

Connection to Lift: PMI = $\log_2(\text{Lift})$

A19: MDL — Minimum Description Length

MDL is a principle for model selection: the best model is the one that gives the shortest total description.

Idea: Imagine you need to send a dataset to a friend. You could either:

1. Send the raw data (very long!)
2. Send some rules + exceptions (potentially shorter)

Trade-off:

- More rules = shorter data description, but longer rule description
- Fewer rules = shorter rule description, but longer data description

Optimal: Find the sweet spot that minimizes total length.

Why it helps: MDL naturally prevents overfitting (memorizing noise) because noise is expensive to describe.

Quick Reference Table

Symbol	Name	Meaning
$H(X)$	Entropy	Average surprise/uncertainty in X
$H(B\|A)$	Conditional entropy	Uncertainty in B after knowing A
$p(x)$	Probability	Chance of x happening (0 to 1)
$\log_2$	Log base 2	"What power of 2 gives this?"
$\sum$	Sum	Add up all terms
$\binom{n}{k}$	Combinations	Ways to choose k from n
k	Antecedent size	Number of conditions in rule
n	Column count	Total number of columns

Metric	Formula Intuition	What It Measures
Support	How often does this apply?	Frequency of rule in data
Confidence	When it applies, how accurate?	Reliability of rule
Lift	Better than random?	Improvement over guessing
Info Gain	How much uncertainty removed?	Value of the information
PMI	Related or independent?	Strength of association

Table 6: Quick reference for rule quality metrics